

Current Numerical Results

13 March 2026

Purpose

This note collects the current benchmark and gDMR results in one short place. The Heston case is kept as the published benchmark. For the gDMR runs, where no finite-difference benchmark is available in the project, the LSMC direct estimator is used as the working reference.

1. Paper Benchmark: Heston Bermudan Put

This case is included to check the implementation against the published Heston benchmark. It is the only case here with a finite-difference reference value.

Table 1: Benchmark setup for the paper-style Heston case.

Quantity	Value
Model	Heston
Option	Bermudan put
S_0	10
K	10
T	1
r	0.02
v_0	0.15
κ	5
θ	0.16
η	0.9
ρ	0.1
Exercise dates	12
LSMC paths	500,000
LSMC steps M	1,000
Hybrid paths N	10,000
Hybrid low paths N_{low}	10,000
Asset grid resolution	512
Finite-difference reference	1.4507

Table 2: Benchmark results for the paper-style Heston case.

Estimator	Published value	Current run	Run – FD	Run – published
LSMC direct	1.4494	1.458896	0.008196	+0.009496
LSMC low	1.4487	1.452544	0.001844	+0.003844
Hybrid direct	1.4530	1.452988	0.002288	-0.000012
Hybrid low	1.4529	1.445493	0.005207	-0.007407

2. gDMR Baseline: $\delta_1 = \delta_2 = 0.50$

This is the current default gDMR run. Relative errors in this section are measured against the LSMC direct result.

3. gDMR Equal-Delta Case: $\delta_1 = \delta_2 = 0.75$

This run keeps the same numerical configuration as the baseline case and changes only the two elasticity parameters to $\delta_1 = \delta_2 = 0.75$.

4. Comparison of the Two gDMR Cases

The next table compares the baseline gDMR case and the equal-delta case under the same numerical setup.

5. Equal-Delta Strike Tests

For these runs, the model parameters and numerical settings are kept fixed at the equal-delta case, and only the strike is changed to produce ITM, ATM, and OTM puts.

Table 3: Setup for the current gDMR baseline run.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100
K	100
T	1
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
ρ_{12}	0.20
ρ_{13}	0.10
ρ_{23}	0.10
δ_1	0.50
δ_2	0.50
Exercise dates	100
Euler steps M	600
LSMC paths	1,000,000
Hybrid paths N	30,000
Hybrid low paths N_{low}	30,000
Asset grid points N_S	181
Hermite nodes	64

Table 4: Results for the current gDMR baseline run.

Estimator	Price	Standard error	Relative error
LSMC direct	6.384535	0.007943	reference
LSMC low	6.387249	0.007939	0.04% vs LSMC direct
Hybrid direct	6.421758	0.000652	0.58% vs LSMC direct
Hybrid low	6.265070	0.045711	1.87% vs LSMC direct

Table 5: Additional comparison for the current gDMR baseline run.

Quantity	Value
Hybrid low relative error vs LSMC low	1.91%

Table 6: Setup for the equal-delta gDMR run.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100
K	100
T	1
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
ρ_{12}	0.20
ρ_{13}	0.10
ρ_{23}	0.10
δ_1	0.75
δ_2	0.75
Exercise dates	100
Euler steps M	600
LSMC paths	1,000,000
Hybrid paths N	30,000
Hybrid low paths N_{low}	30,000
Asset grid points N_S	181
Hermite nodes	64

Table 7: Results for the equal-delta gDMR run.

Estimator	Price	Standard error	Relative error
LSMC direct	6.653560	0.007822	reference
LSMC low	6.655062	0.007819	0.02% vs LSMC direct
Hybrid direct	6.696152	0.000375	0.64% vs LSMC direct
Hybrid low	6.545257	0.045080	1.63% vs LSMC direct

Table 8: Additional comparison for the equal-delta gDMR run.

Quantity	Value
Hybrid low relative error vs LSMC low	1.65%
1% direct target met	yes

Table 9: Comparison between the two gDMR parameter cases.

Estimator	$\delta = 0.50$	$\delta = 0.75$	Change
LSMC direct	6.384535	6.653560	+0.269025
LSMC low	6.387249	6.655062	+0.267813
Hybrid direct	6.421758	6.696152	+0.274394
Hybrid low	6.265070	6.545257	+0.280187

Table 10: Strike-test setup for the equal-delta gDMR runs.

Case	S_0	K
ITM put	100	110
ATM put	100	100
OTM put	100	90

Table 11: Strike-test results for the equal-delta gDMR runs.

Case	Strike	LSMC direct	LSMC low	Hybrid direct	Hybrid low	Hybrid direct rel. error	Hybrid low rel. vs LSMC direct	1% target met
ITM put	110	12.735850	12.734082	12.767870	12.634384	0.25%	0.80%	yes
ATM put	100	6.653560	6.655062	6.696152	6.545257	0.64%	1.63%	yes
OTM put	90	2.743578	2.734472	2.791426	2.668242	1.74%	2.75%	no

Table 12: Additional relative errors for the equal-delta strike tests.

Case	Hybrid low rel. vs LSMC low	LSMC low rel. vs LSMC direct
ITM put	0.78%	0.01%
ATM put	1.65%	0.02%
OTM put	2.42%	0.33%